Guidelines for Meshing Impeller for CFD Analysis

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1. Purpose

The document describes a guideline to realize a mesh of impeller with a quality level sufficient to adequately capture fluid-dynamic phenomena, minimizing the number of elements of the computational grid.

A specific example using CFX Turbogrid of ANSYS environment is described in the Annex.

2. Scope

This procedure is applicable to any vaned channel of centrifugal pumps, rotoric or statoric, radial, mixed or axial flow, single or double suctions, overhung or between bearing solution.

The procedure applies to the Nova Milanese and La Spezia plants.

3. Abbreviations

CFD		*Computational fluid Dynamics*
D1	*[mm or in]*	*impeller eye diameter*
D2	*[mm or in]*	*maximum external diameter of the impeller*
Dh	*[mm or in]*	*shaft impeller diameter*
Eta	*[%]*	*efficiency*
FVM		*Finite Volume Metod*
G		*growth ratio of the cell height*
H	*[m]*	*head*
LE		*leading edge of blades impeller*
N		*number cell of boundary layer*
Nq or Ns		*specific speed of the pump*
PS		*pressure side of blades impeller*
Q	*[m^3/h]*	*flowrate*
Ra	*[µm]*	*Average Roughness*
SS		*suction side of blades impeller*
TE		*trailing edge of blades impeller*
δ99	*[µm]*	*boundary layer thickness*

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4. General procedure: description

The guideline provides general instructions to design a computational grid of vaned channels starting from a cad model that will be used for CFD analysis. Based on the type of CAD extracted and prepared from SOP-481, one of the following input data is available:

• Airfoils and meridional curves of the vane

• 3D model of the single impeller blade

• 3D model of the complete impeller

It is possible to use different types of grid generators for the impeller mesh; depending on the available type of cad, the following different approaches will be adopted to create the mesh.

A general discussion is presented below, with the aim to provide the user with the necessary information and recommended criteria for meshing the vaned channels using general purpose tools.

1. Numerical Discretization techniques: FVM

Analytical solutions of the Navier-Stokes equations are able to solve only the simplest flows under ideal conditions. To obtain solutions for real flows, a numerical approach must be adopted whereby the equations are replaced by algebraic approximations which may be solved using a numerical method

The techniques of Finite Volume Method is the "classical" or standard approach used most often in commercial software and research codes. The governing equations are solved on discrete control volumes FVM (Figure1) recasts the PDE's of the N-S equation in the conservative form and then discretize this equation. This guarantees the conservation of fluxes through a particular control volume. Though the overall solution will be conservative in nature there is no guarantee that it is the actual solution. Moreover, this method is sensitive to distorted elements which can prevent convergence if such elements are in critical flow regions.

The following guidelines provide a series of criteria and settings developed over the years through a persistent series of analyzes and comparisons that guarantee an adequate degree of reliability of the results obtained compared to the experimental data measured in the field.

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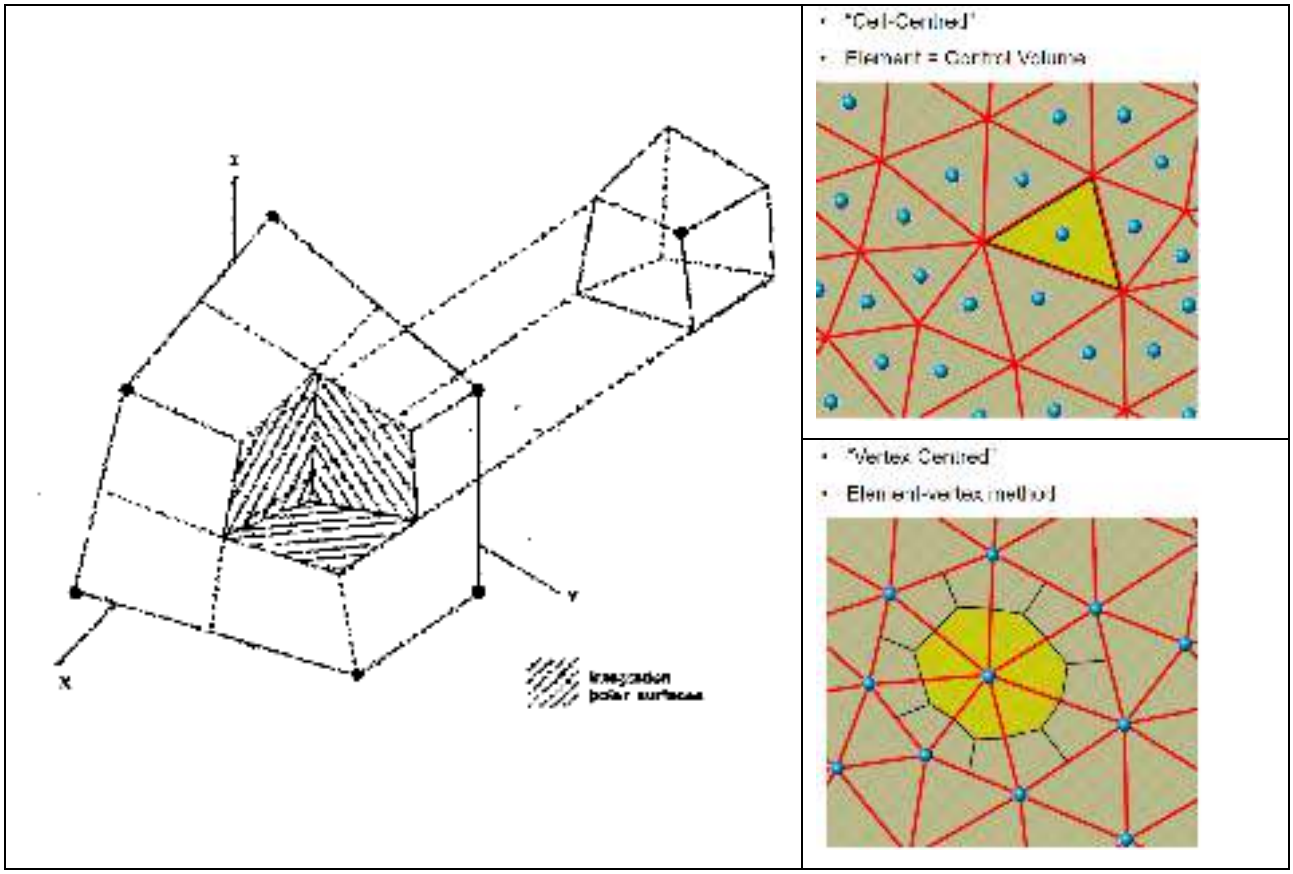

Figure 1 : FVM, method Cell-Centred and method Vertex-Centred

2. Element types of cell mesh

Figure2 shows different kinds of mesh element, the appropriate choice and placement of these elements that discretize the fluid domain depends on which fluid dynamic phenomenon you want to model and with what degree of accuracy.

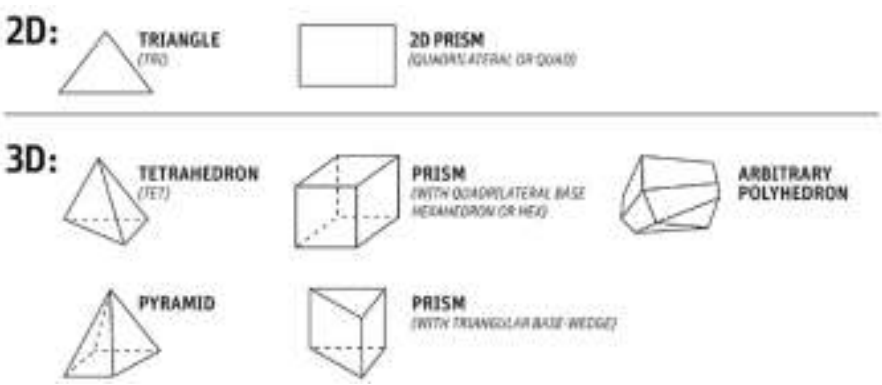

Figure 2 : type mesh element.

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Prismatic elements are the best in terms of quality, but they cannot always be used due to the irregular shapes of the geometry, however it is recommended to always use them to simulate the boundary layer.

3. Typology of mesh: structured and unstructured grid

Mesh elements can be generated according to two different types of logic (Figure3):

Non structured:

Any agglomeration of elementary elements, organized or not. Most of the meshes in this category are hybrid meshes using tetrahedrons and prisms.

Structured:

An organized agglomeration of hexahedrons.

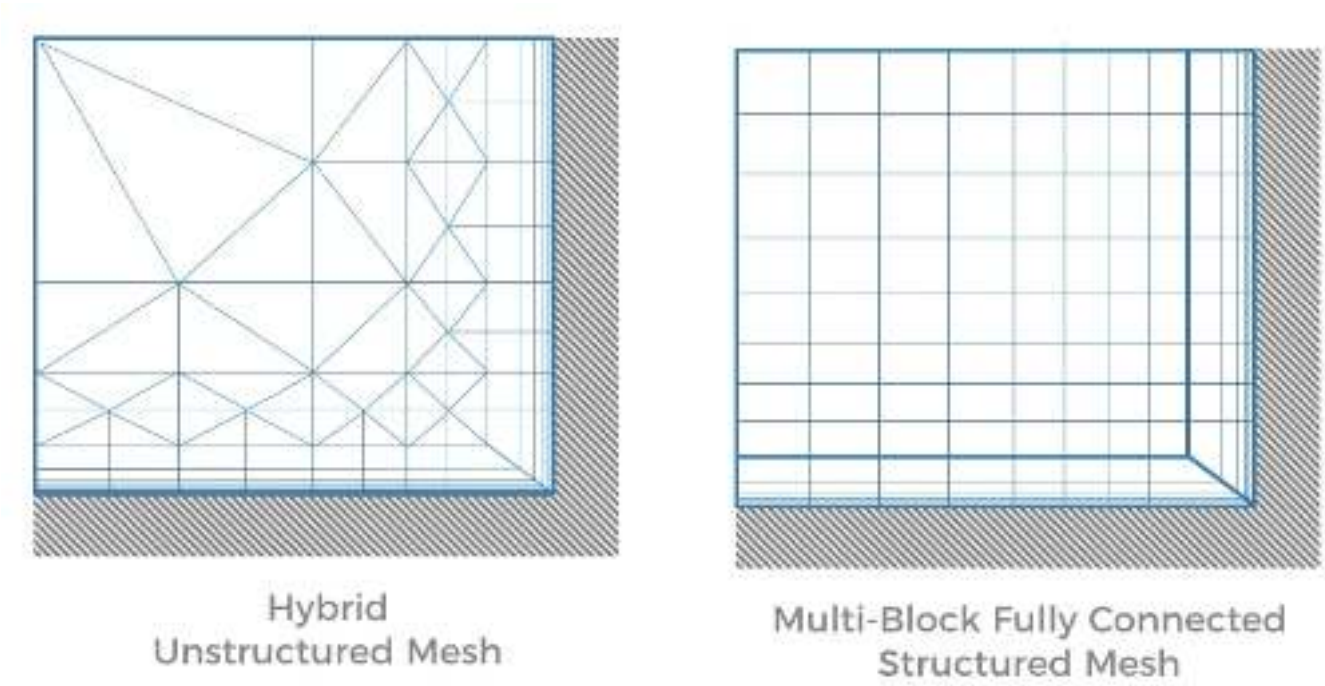

Figure 3 : structured and unstructured grid.

The choice between the two types of mesh depends on the complexity of the geometry, the accuracy of the numerical result we want to obtain and the type of mesh tool available.

Although hexahedral elements are best at resolve and transmitting information between adjacent cells, they are only applicable for simple geometries; for complex shaped geometries it is necessary to switch to an unstructured grid.

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In general, if you have "Airfoils and meridional curves of the vane" available, it is possible to create a Structured Mesh (for example using the ANSYS Turbogrid tool), while with "3D model of the single impeller blade" and "3D model of the complete impeller" it is easier to create a Hybrid Unstructured Mesh with a generic tool (for example ICEM by ANSYS).

4. Topology of mesh: structured C grid and O grid

There are various mesh topologies in the literature that are best suited for certain geometries and fluid dynamic phenomena.

Two examples are shown in the Figure4, the "C" type mesh and the "O" type mesh, these are well suited for meshing hydraulic machines:

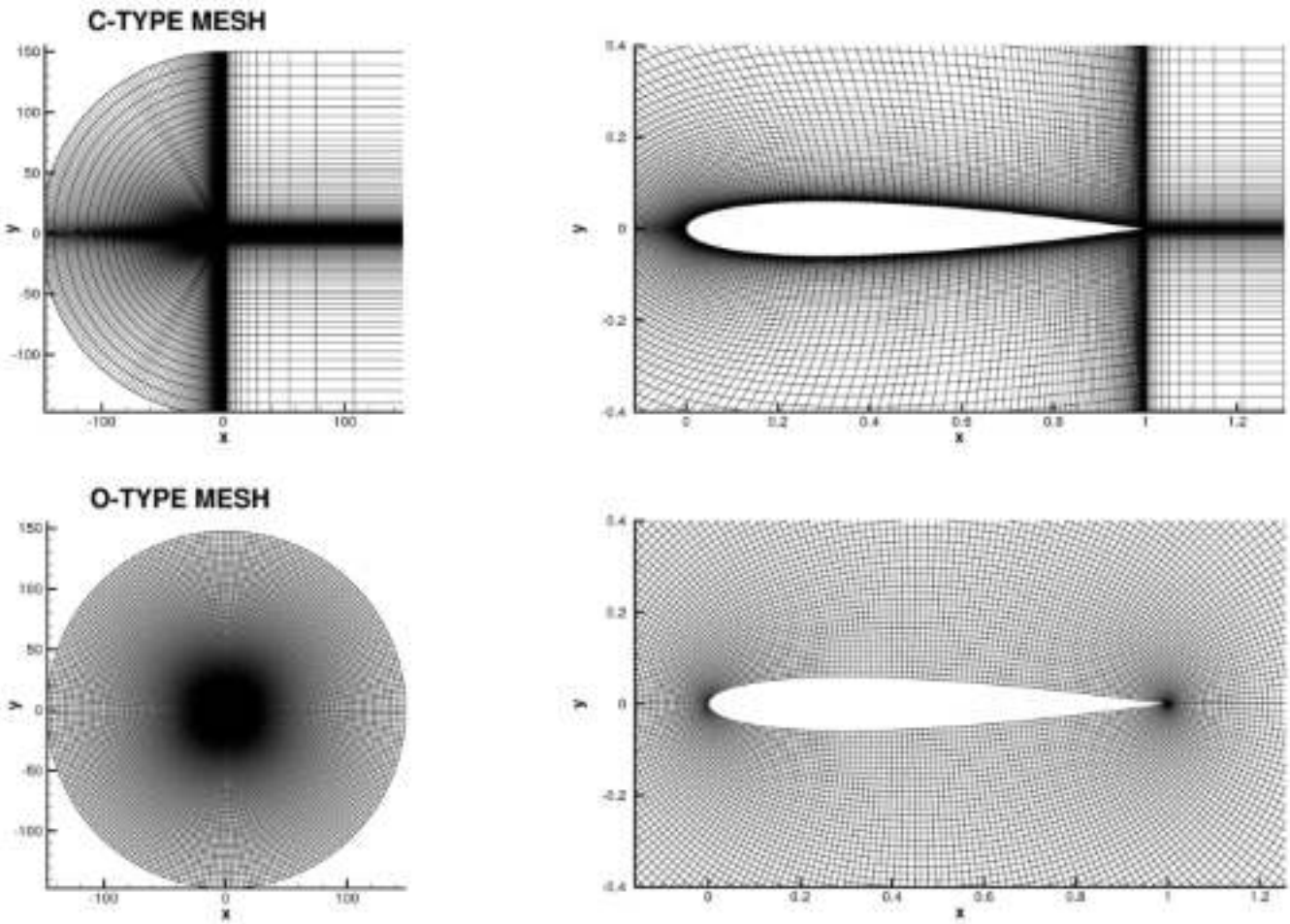

Figure 4 : structured C grid – O grid.

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The "O" type mesh is the best because it discretizes and captures the phenomena in proximity in the LE and TE well and optimizes the distribution of elements in the fluid domain.

It often happens that during "O" type meshing, due to the reduced space between the TE and the interface, the generated cells are distorted and of poor quality; to overcome this problem it is necessary to use a "C" type mesh.

5. Number of grid elements and simulation time

In general about 1 million of elements are necessary for the entire impeller (range 600 000 – 1 400 000 number of elements is a good quantity).
The final number of mesh elements varies from case to case and depends on the degree of refinement you want to give, since the mesh must be refined in zones with high velocity gradients, so near the vane LE - TE and near wall.
Sometimes we are forced to increase the number of elements when the quality of the mesh is not good, in this phase it is important to distribute the mesh elements properly because even if it is true that a greater number leads to a better quality of the solution but on the other hand it requires more simulation time.

6. Inlet and outlet of mesh

Inlet and outlet of the impeller mesh are the regions assigned for the interface with the stator meshes, their distance from the blades must be as small as possible to make the space between impeller and stator domain larger enough to guarantee a good mesh quality on the stator domain without any worsening of the mesh quality near to the LE and TE of the impeller.
To have conformity in the interfaces between the stator and rotor meshes, they must have approximately the same number of elements and approximately the same distribution.
If the cells are particularly distorted in terms of orthogonality and aspect ratio, it is possible that an incorrect transmission of information between the two meshes may occur.

7. Boundary Layer: Modeling Flow Near the Wall and Height of first cell

Near the solid wall, the modeling of the fluid boundary layer requires particular attention, since in this region strong gradients of the fluid dynamic variables calculated by the CFD solver occur.
Experiments and mathematical analyzes have shown that the region near the wall can be divided into two layers:

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-in the innermost layer, close to the wall, the so-called "viscous sublayer", the flow is almost laminar and the (molecular) viscosity plays a dominant role in the momentum and heat transfer.

- in the layer furthest from the wall, in the logarithmic layer, turbulence dominates the mixing process.

The Figure5 below illustrates these subdivisions of the near-wall region and an example of discretization of boundary layer.

Note as a fine grid resolution is required to accurately resolve the gradient of velocity.

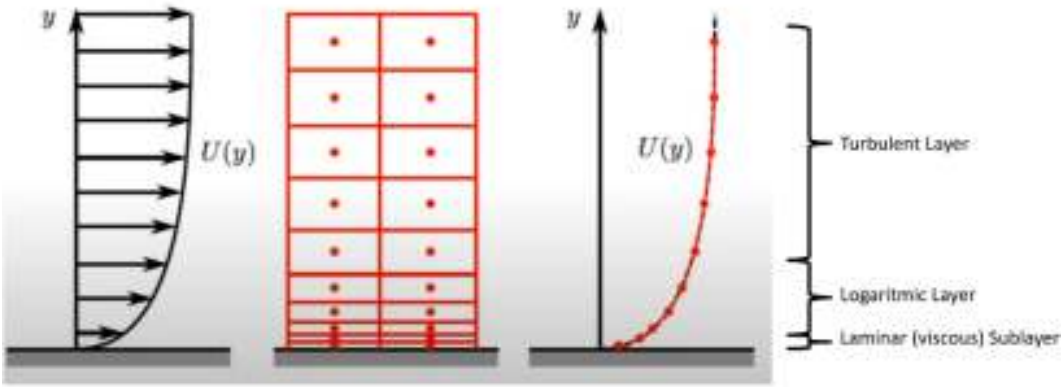

Figure 5 : boundary layer model.

The boundary layer can be either modeled using "wall functions" (empirical functions obtained from experiments and DNS simulation that are fitted to the observed behavior close to the wall) or simulated by imposing the smallest possible height of the first cell and an adequate growth ratio.

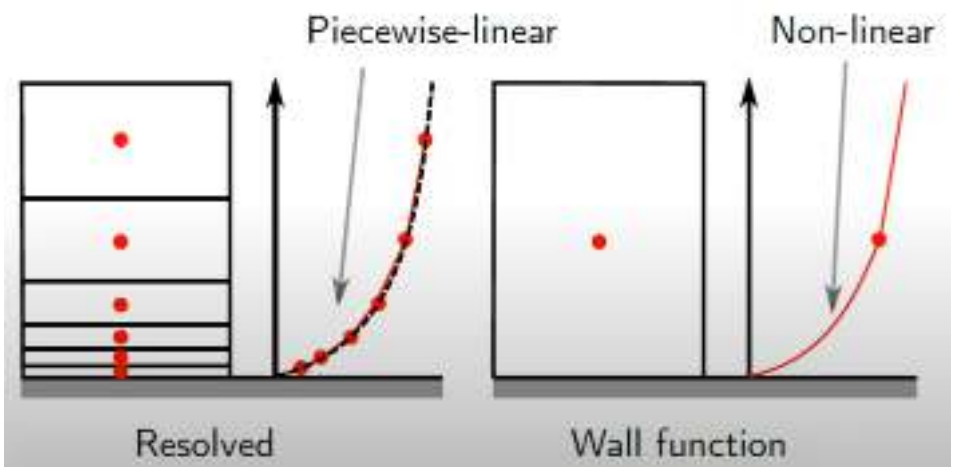

Figure 6 : boundary layer model, resolved and wall function.

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Since different stationary/unstationary velocity fields arise near the wall in the fluid domain, it is not possible at present to adapt the height from the first cell to the various needs, we are therefore forced to a compromise value.

	Height of first cell [μm]	
	Wall function	Low Reynolds method
Smooth wall	200-300	1
Wall with roughness	5*Ra	/

The following Figure7 shows the relative height value of the first cell depending on the roughness Ra (Averaged Roughness) in American and Metric system.

Approximate Surface Roughness Conversion Chart					
Roughness Grade Numbers	American System		Metric System		Height first cell mesh (μm)
	Ra (μin)	RMS (μin)	Ra (μm)	RMS (μm)	
N12	2000	2200	50	55	64
N11	1000	1100	25	27.5	32
N10	500	550	12.5	13.75	16
N9	250	275	6.3	9.13	8
N8	125	137.5	3.2	3.52	4
N7	63	69.3	1.6	1.76	2
N6	32	35.2	0.8	0.88	1
N5	16	17.6	0.4	0.44	1
N4	8	8.8	0.2	0.22	1
N3	4	4.4	0.1	0.11	1
N2	2	2.2	0.05	0.055	1
N1	1	1.1	0.025	0.035	1

Figure 7 : heigh first cell mesh with Ra.

For further details consult [2]

8. Boundary Layer: Grow rate, boundary layer and spanwise blade elements

Once the method that defines the thickness of the first cell has been chosen, which as mentioned in the previous paragraph we conventionally assume to be equal to:

Height of first cell for “Wall functions”: 200 micrometers.

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Height of first cell for “Low Reynolds method “: 1 micrometer

It is necessary to define the growth ratio of the cell height G and the number of cells N.
Two good rules are listed below (Figure8):

1) the boundary layer thickness δ_{99} must be contained within the inflation layers.

2) Increase the N number of layers keeping recommended G in the range 1.05<G<1.4.

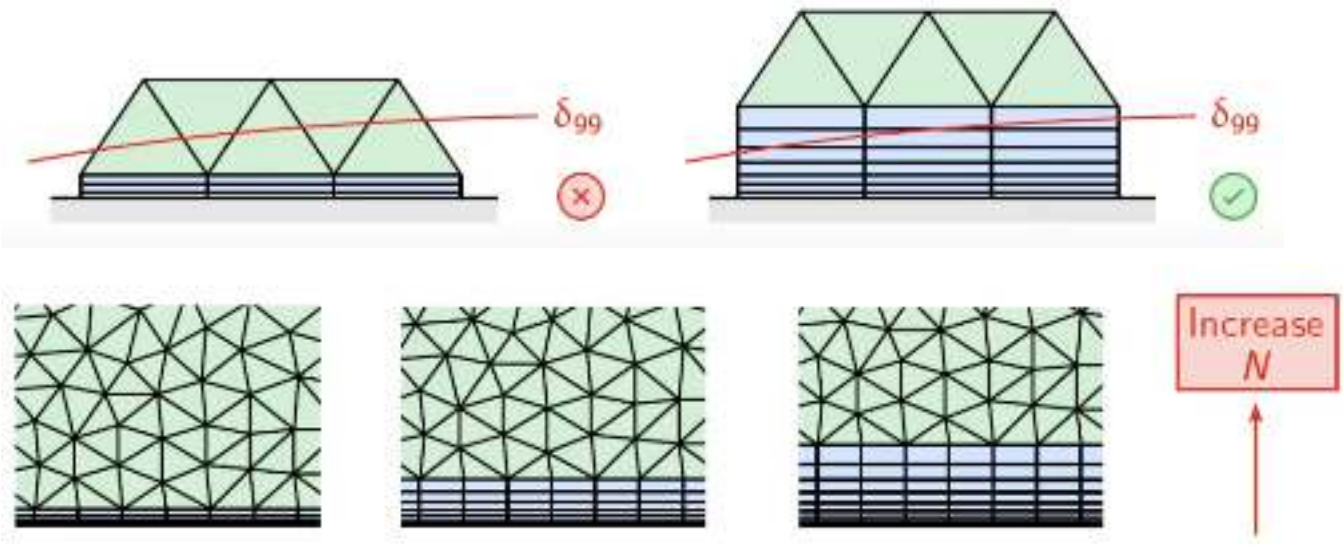

Figure 8 : Good rules for G and N

Since the thickness of the boundary layer is not known regardless, also in this case the following conventional values are assumed:
N= 8-16
G=1.4

For the number of elements in the spanwise blade (Figure9) typical values are as follows:
Total elements on span = range 30-40
Constant elements = about 22-24
Boundary layer elements = range 8-16

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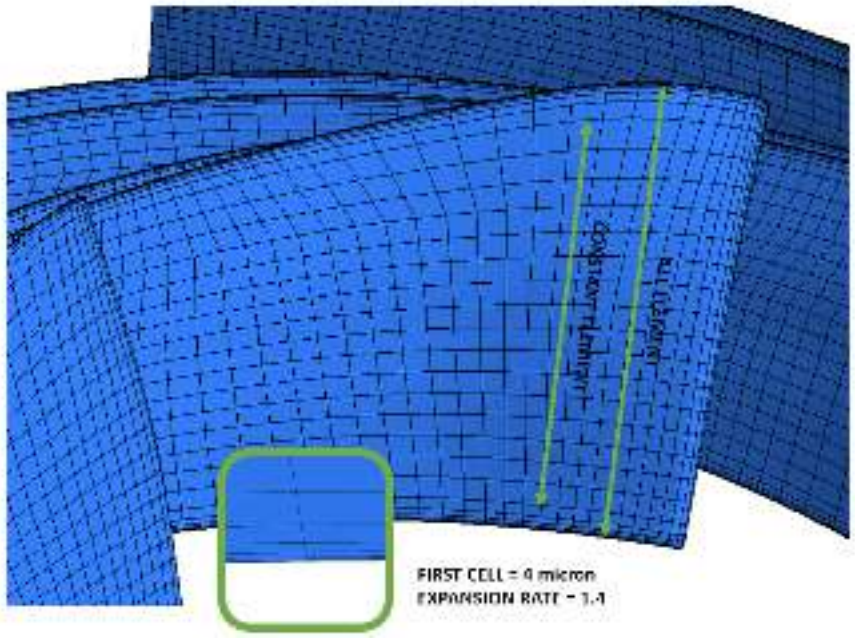

Figure 9 : example elements in the spanwise blade.

9. LE TE mesh

The fluid region close to the LE is critical as it is the region of incidence of the flow on the blade, therefore it must have an adequate degree of discretization, generally 8-20 elements arranged on the LE are sufficient to follow its curvature well.

The fluid region close to the TE is also critical since downstream of it a turbulent wake is created whose size and intensity are mainly linked to the thickness of the blade in this section, generally 8-20 elements arranged on the TE are sufficient to follow its shape well (Figure10).

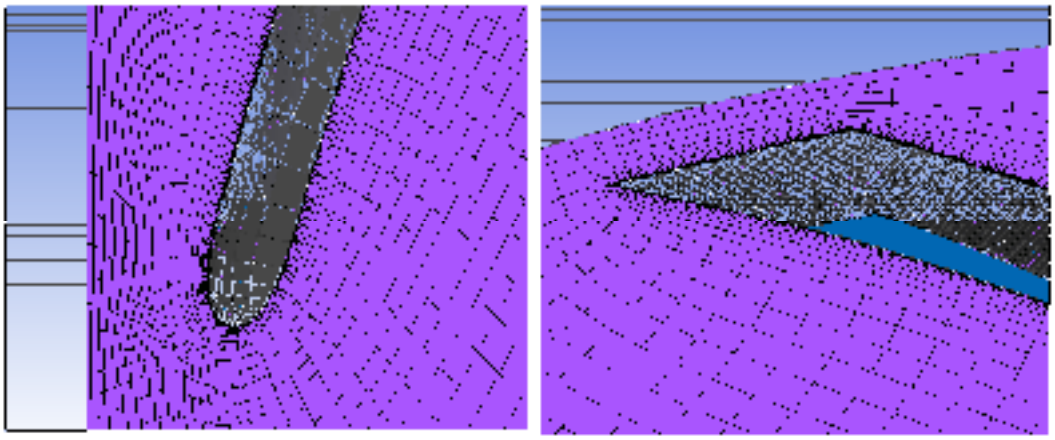

Figure 10 : refinement mesh at LE and TE

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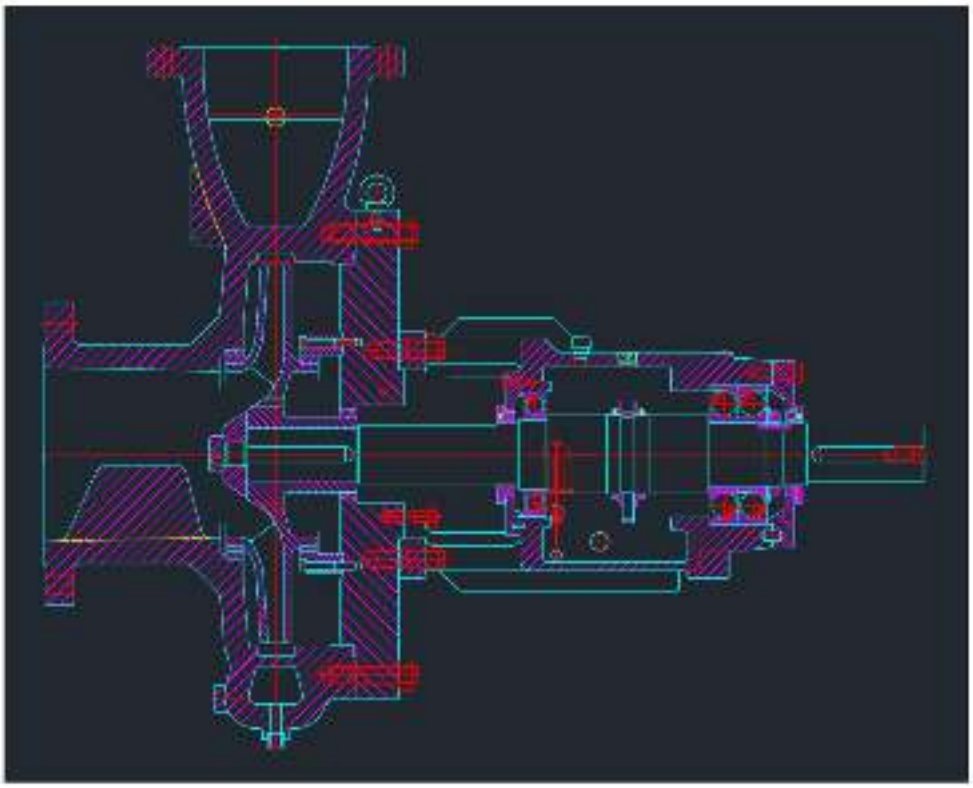
10. Impeller with balance holes

If the impeller has balancing holes, the approach that must be adopted to take their presence into account in the three types of extracted CAD is the following:

- with "Airfoils and meridional curves of the vane" and "3D model of the single impeller blade" it is not possible to manage the presence of holes within a single impeller mesh (the impeller mesh is typically structured), it is necessary to create another dedicated mesh for the holes (typically with an unstructured mesh) and in a second phase create the interfaces between the two domains so that they can communicate with each other (Figure 11).



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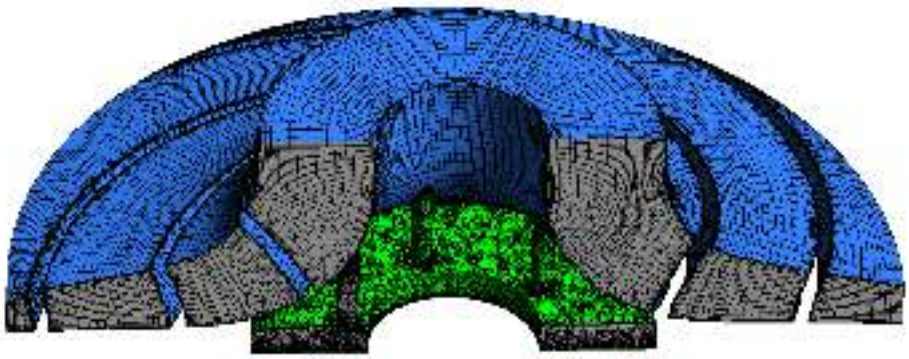

Figure 11 : dedicated mesh for the balance holes.

- with "3D model of the complete impeller" however it is possible to create a single impeller mesh including holes, on the other hand, given that the overall geometry is not simple, it is preferable to create the non-structured mesh.

11. Correct periodicity

To avoid problems of stability and convergence of the calculation it is necessary that the "3D model of the single impeller blade" has correct periodicity.

A verification of this condition can be performed by trying to create the "3D model of the complete impeller" starting from a duplication and rotation of the single domain equal to 360° divided by the number of blades impeller and verifying that the interfaces of the individual domains coincide without overlap between them.

12. Mesh quality

The solution of a simulation with FVM strongly depends on the quality of the mesh and the robustness of the solver, so that the results can be considered a valid approximation of the real physical phenomenon the simulation must respect the three fundamental concepts of: consistency, stability and convergence [1].
In the literature there are various criteria for evaluating the quality of the mesh, but the main ones are reported and defined in the following Figure12:

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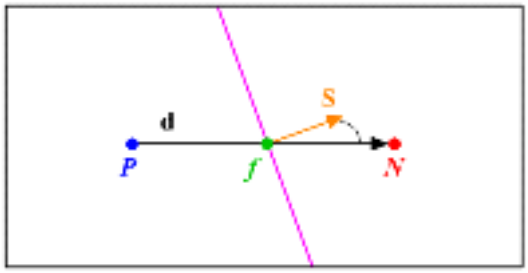
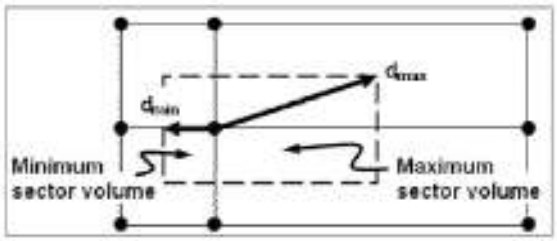


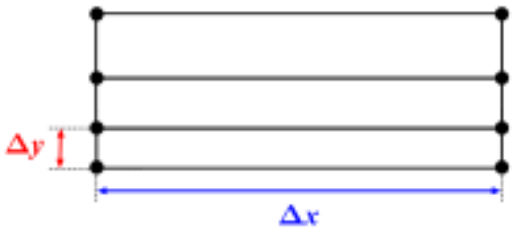
Mesh Orthogonality

Is the angular deviation of the vector S (normal at the diagonal and located at the face center f) and vector d (connecting the two-cell center P and N)



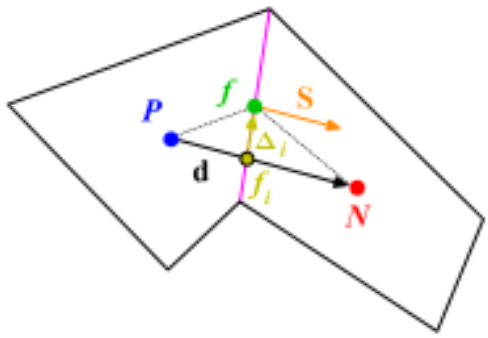
Mesh Expansion Factor

is the ratio of the volume of the larges and smallest sector volumes in element



Mesh Aspect Ratio

Is the ratio of the longest edge length to the shortest edge length of element



Mesh Skewness

Is the deviation of the vector d that connects the two cell P and N, from the face center f.

the deviation vector is represented with Δ and fi is the point where the vector d intersects the face f.

Value of Skewness	Cell Quality
1	degenerate
0.9 — <1	bad (sliver)
0.75 — 0.9	poor
0.5 — 0.75	fair
0.25 — 0.5	good
>0 — 0.25	excellent
0	equilateral

Figure 12 : Criteria for Mesh Quality

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The acceptability ranges for the different defined criteria are listed below:

Mesh Orthogonality: the grid lines should be as orthogonal as possible to the walls and within the calculation domain. The elements must not have angles less than 20° or greater than 160° (angles less than 20° compromise not only precision but also convergence).

Mesh Expansion Factor: the size of the elements should not change sharply. The expansion rate of the cells should not change from one cell to the next by more than a factor of 1.1 near walls and 1.5 to 2 in the bulk flow domain; or the cell volume should not increase more than by a factor of 1.5.

Mesh Aspect Ratio: the aspect ratio of the elements should not exceed the factor 10 near walls and 5 in the bulk flow regions. It is particularly important to fulfill these requirements in zones with strong velocity gradients.

Mesh Skewness: the skewness should have at least good or better quality.

No intersections of grid lines (no negative volumes) are to be allowed.

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5. Acceptance criteria

This procedure is applicable for any geometry of impeller; indeed, it could be used for radial, semy-axial and axial impeller, closed, semi-open or open (Table 1).

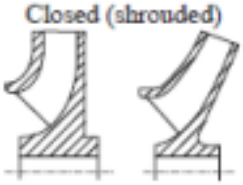
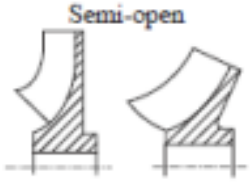
Component or feature	Radial	Semi-axial	Axial
(A) Impeller form: Characteristic: flow direction at impeller exit			
(B) Impeller type	Closed (shrouded) 	Semi-open 	Open

Table 1 : Type of geometry impeller [1]

Furthermore, more in general, this guideline can be used for all the vaned channels, i.e. diffuser of multistage pumps and bowl of vertical pumps (Table 2).

(C) Diffuser Characteristic: flow direction at diffuser inlet	Radial	Semi-axial

Table 2 : Type of geometry diffuser [1]

6. References

[1] Gulich "Centrifugal Pumps" ,4th.ed.

[2] Ansys User Manual

[3] https://www.cfd-online.com/

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Annex A: Creation of mesh impeller using Turbogrid (CFX Ansys®) software

The best practice is to follow every step described in this chapter in the proper order with the aim to reduce the number of operations:

1.1. Extend the curves of meridional channel with four lines, in this way the O-grid mesh will have sufficient space for turn around the blade. So modify the files "hub.curve" and "shroud.curve" in way similar as showed in Figure 13 (in red the lines extended)

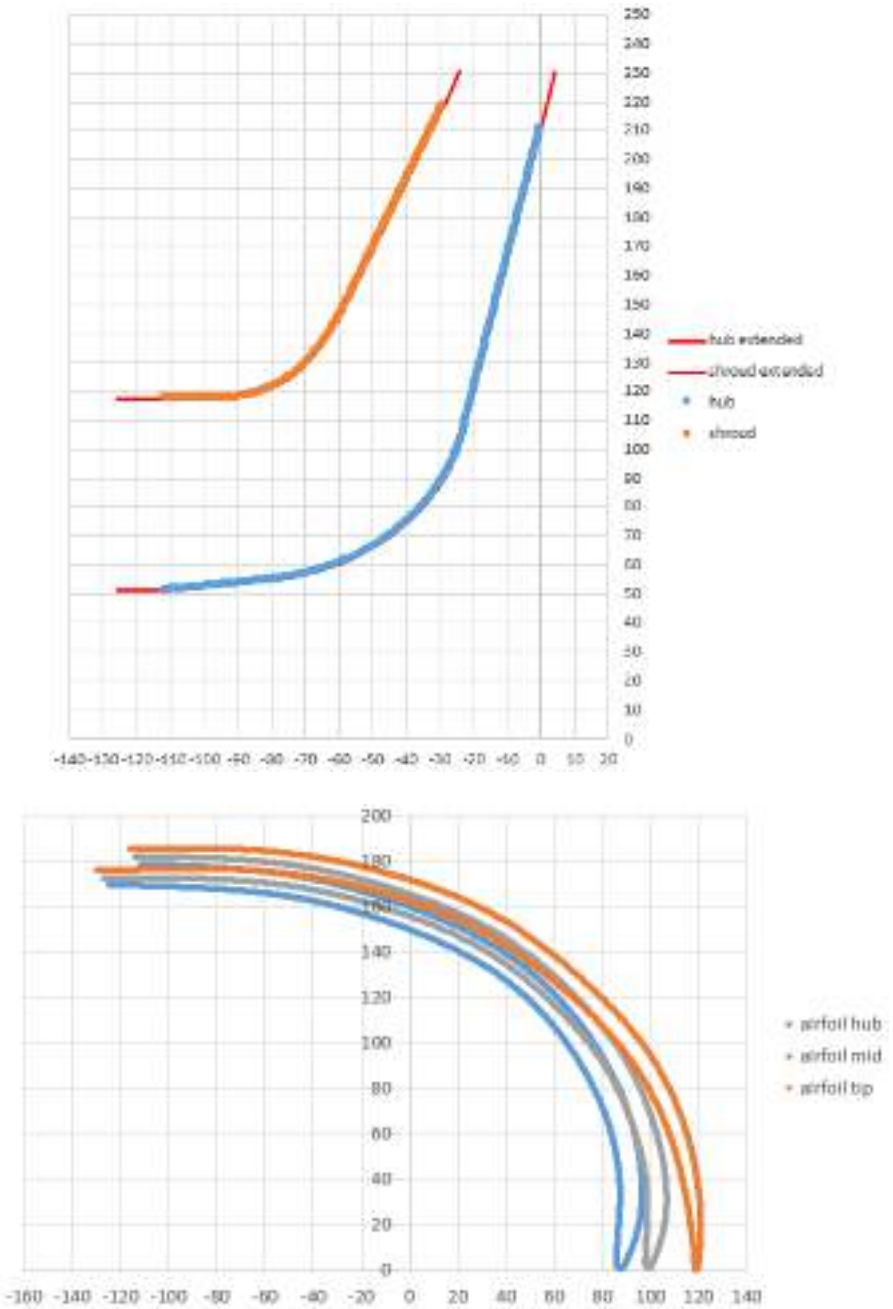

Figure 13 : extension (in red) of meridional channel's curves.

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1.2. Import the geometry files in the geometry section of CFX turbogrid, setting the follow fields:

- length units (for example [m] o [mm]),
- the rotation axis,
- number of blades,
- enable “cut-off or square” in traling edge (TE) section to get the TE of blades squared.

Accepting the initial setting the geometry of impeller will be showed in the graphic window as showed in Figure 14.

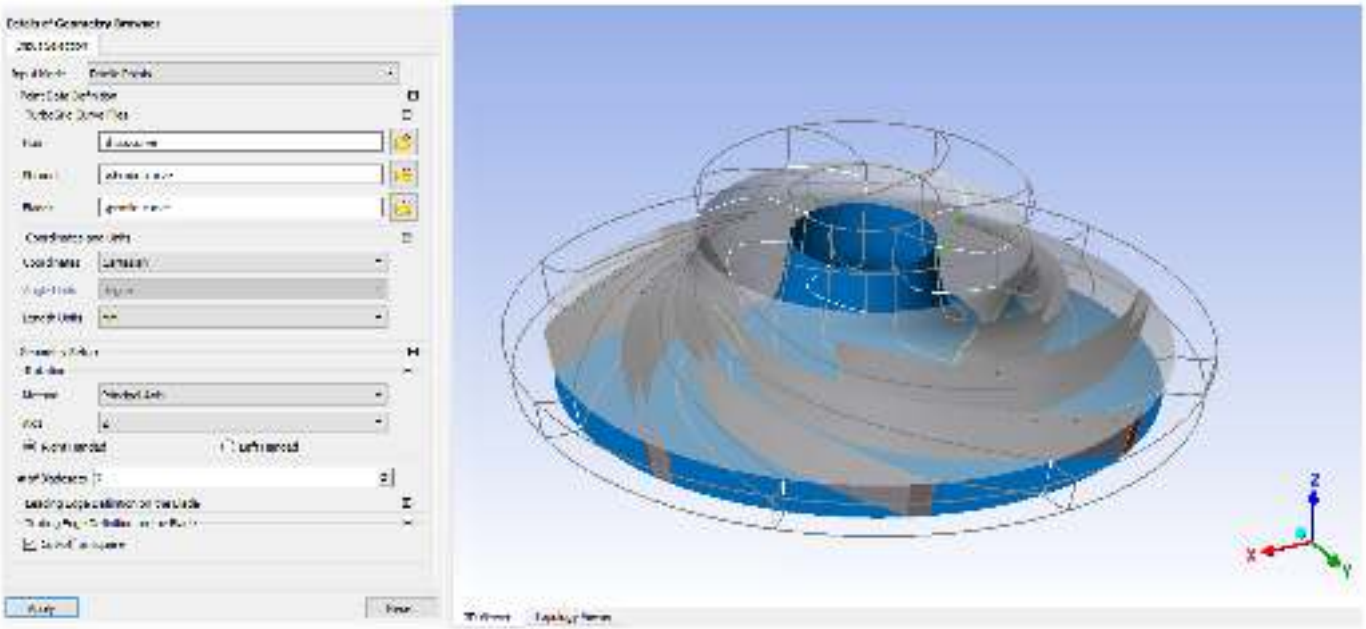

Figure 14 : Setting and import geometry files.

1.3. Set in the model tree “Topology set” active the command “Suspend Object Updates”, in this way all the operation and modify done will not be updates every times.

1.4. Setting the limit of interface P1,P2,P3,P4 in according to dimension of leakage interfaces (Figure 15).

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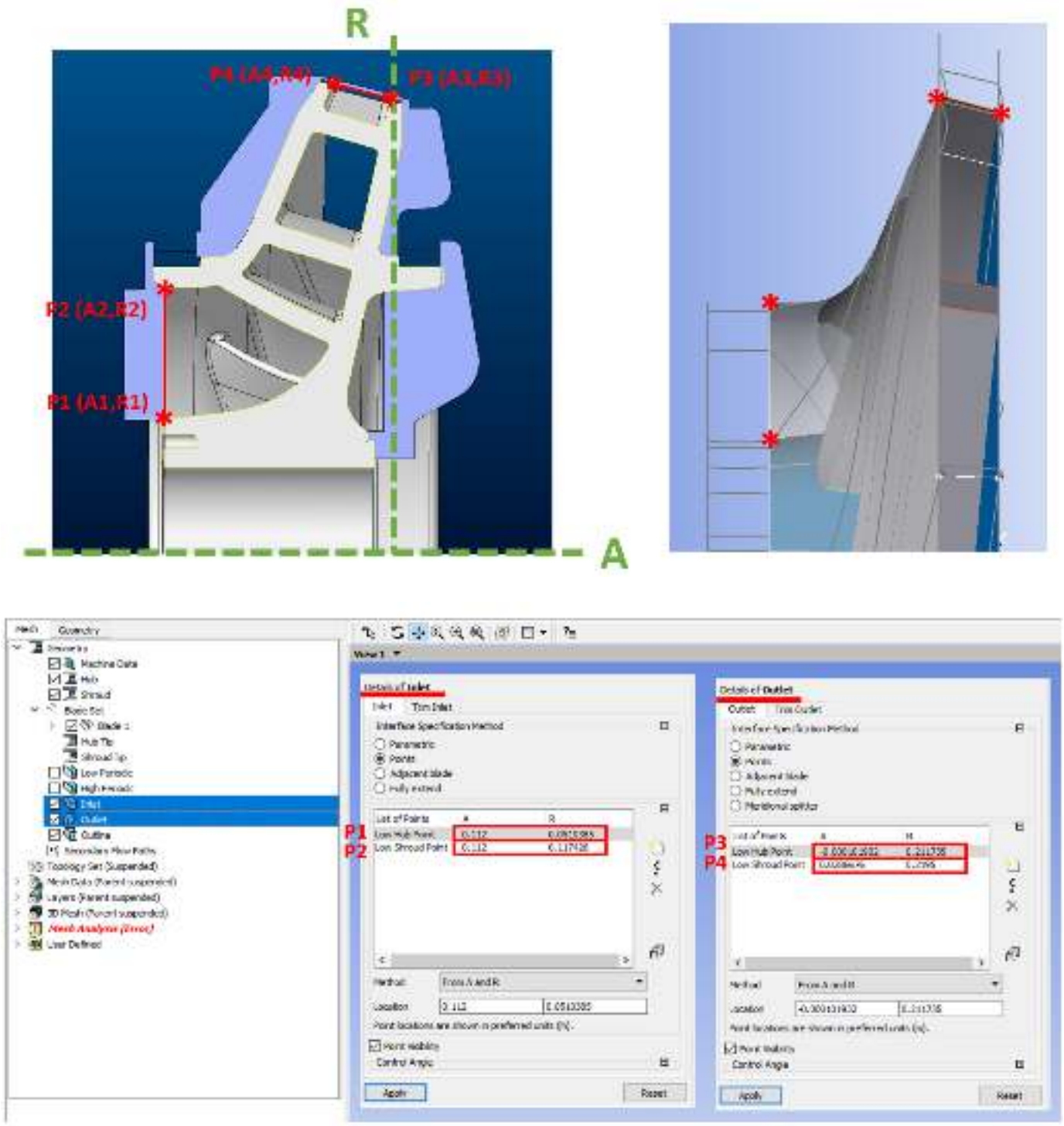

Figure 15: setting the limit of interface in according to dimension of leakage interfaces.

The difference can get the main geometric parameters of meridional channel's impeller as showed in Table 3:

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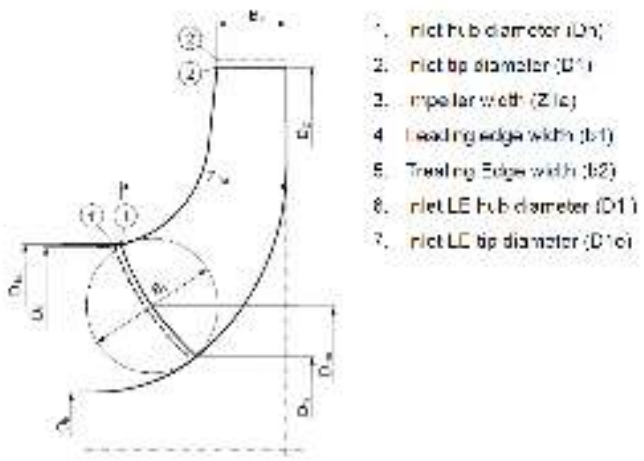
Relation between coordinate points P1,P2,P3,P4 and main geometric parameters of the meridional channel	
R2-R1=B1; *((A4-A3)^2+(R4-R3)^2)^0.5=B2;* *R1*2=Dh* *R2*2=D1a* *((R4+R3)/2)*2)-X=D2m* *|A1-A3|=X2h*	

Table 3 : Relation between extreme points mesh and main geometric parameters of the meridional channel

Inside the table X represent a dimension added necessary that allow boundary layer of leakage turn through the direction of boundary layer of impeller, so for optimize communication of interface.

Since for the norm API 610 the ratios between Dcw/D2 ≥106% (relative combination impeller-volute) and D3/D2 ≥ 103% (relative combination impeller-diffuser) are constrained, the gap between two component of geometry is stringent.
Is good practice setting X as written in follow Table 4:

Pump with	Norm API 610	X
combination impeller-volute	Dcw/D2 ≥106%	((Dcw-D2)/D2)/6=1%
combination impeller-diffusor	D3/D2 ≥ 103%	((D3-D2)/D2)/6=0.5%

Table 4 : Dimension of parameter X

If from this table, the result gotten is X≥ 0.5mm is preferable to limit this value at 0.5mm, in this way the modelled disks of the impeller are not too oversized (Figure 16).

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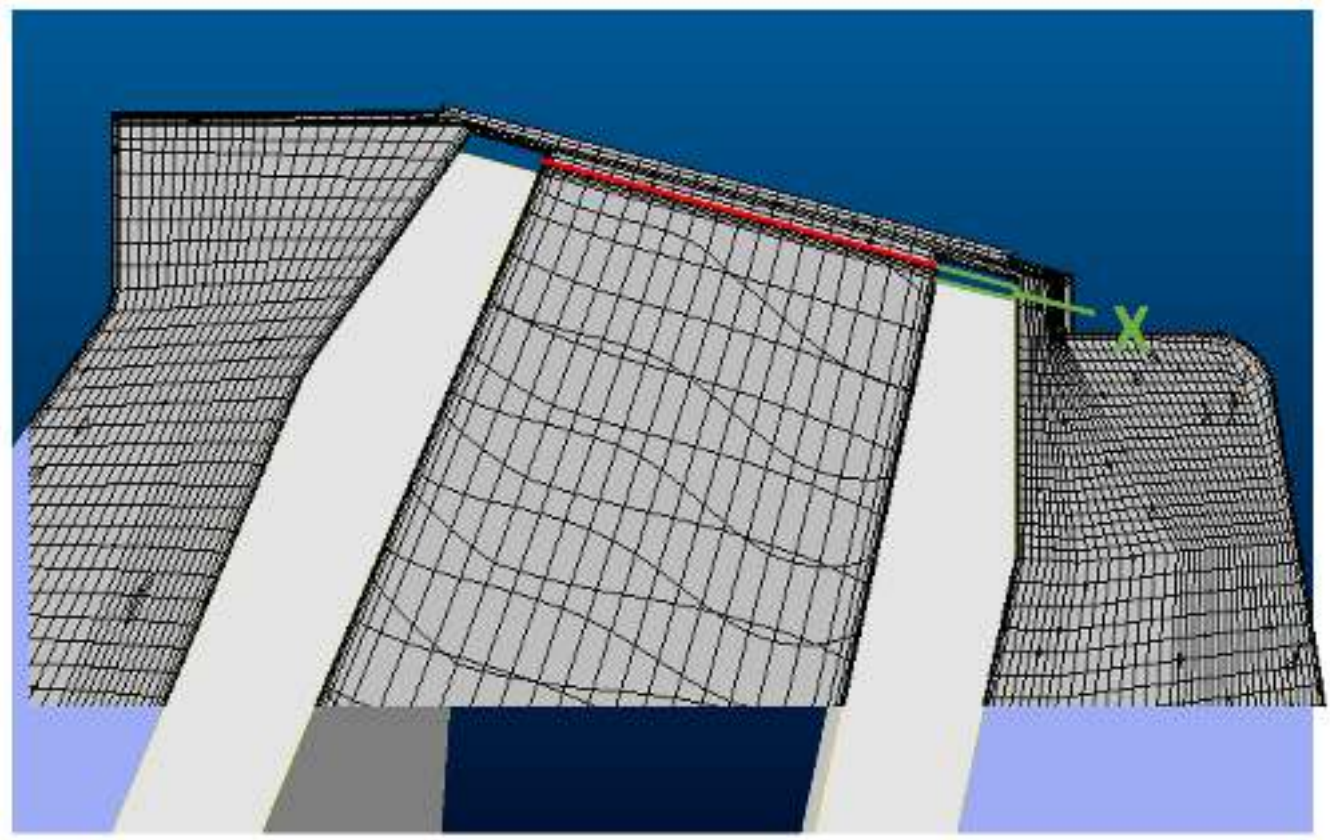

Figure 16: dimension of parameter X

Instead, the axial gap between LE and inlet interface of impeller is preferable extend for all the horizontal length available.

1.5. Set in "Mesh Data" in the section "Mesh Size" the follow fields (Figure 17):

- *Method: Target Passage Mesh Size*
- *Target: 200 000*

This is the number of elements for every vane of impeller, since the number of impeller is sometimes 3, 5 or 7 blades, the global mesh could be respectively of 600 000, 1 000 000 or 1 400 000 number of elements.

- *Boundary layer refinement control, method: First Element Offset*
- *Boundary layer refinement control, offset: 4 [micron].*

This dimension is the height of the first cell of meh (so, in the setting of CFX this value correspond at 8 micron of roughness).

- *Target maximum expansion rate: 1.4*

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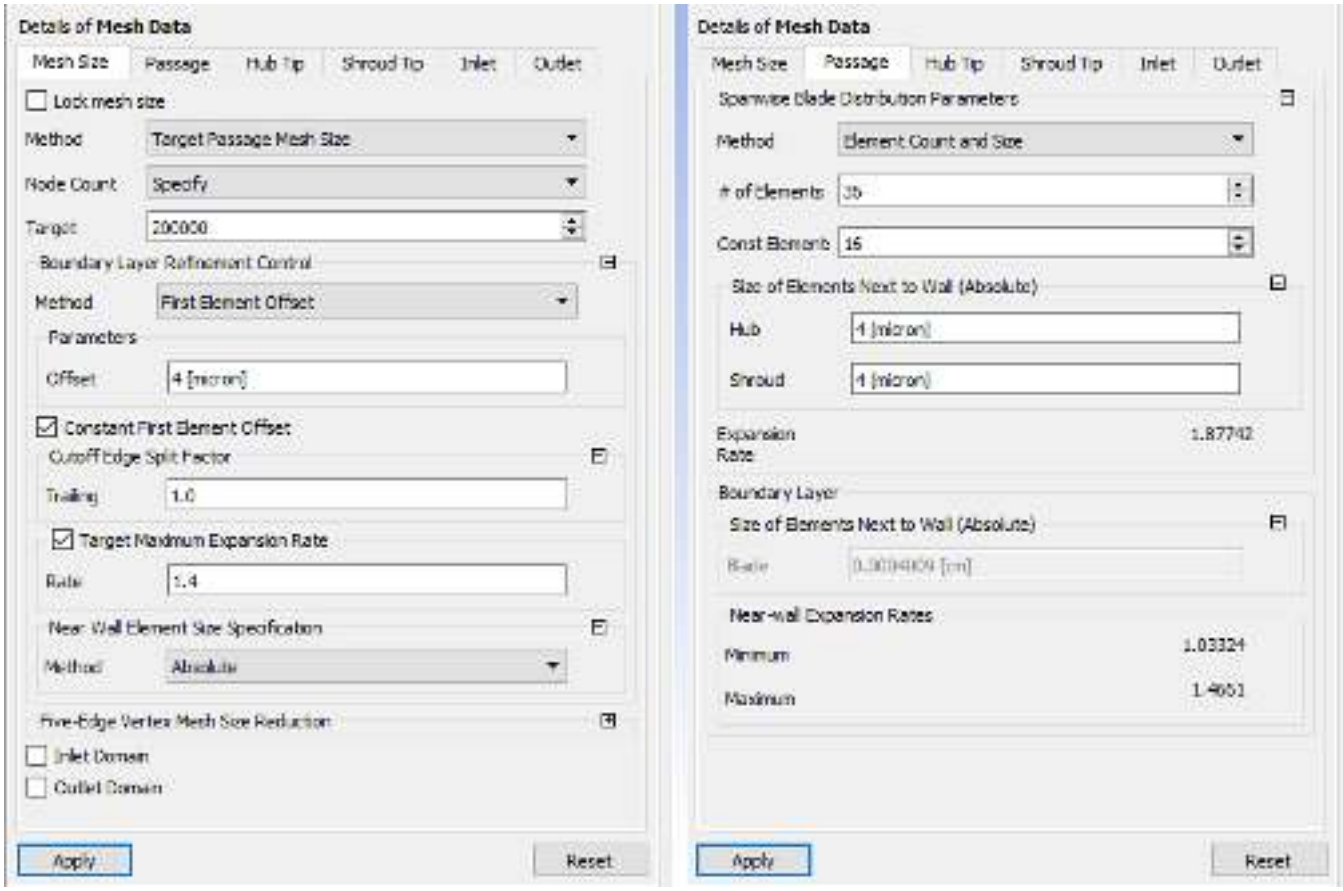
Instead set in the section “Passage” the follow fields:

- *Spanwise Blade Distribution Parameters, method: Element Count and Size*
- *Spanwise Blade Distribution Parameters, # of Elements: 36;* these are all the elements of cell along the span of blade’s impeller
- *Spanwise Blade Distribution Parameters, Cost Elements: 16*
- *Size of Elements Next to Wall, hub: 4 [micron]*
- *Size of Elements Next to Wall, shroud: 4 [micron]*



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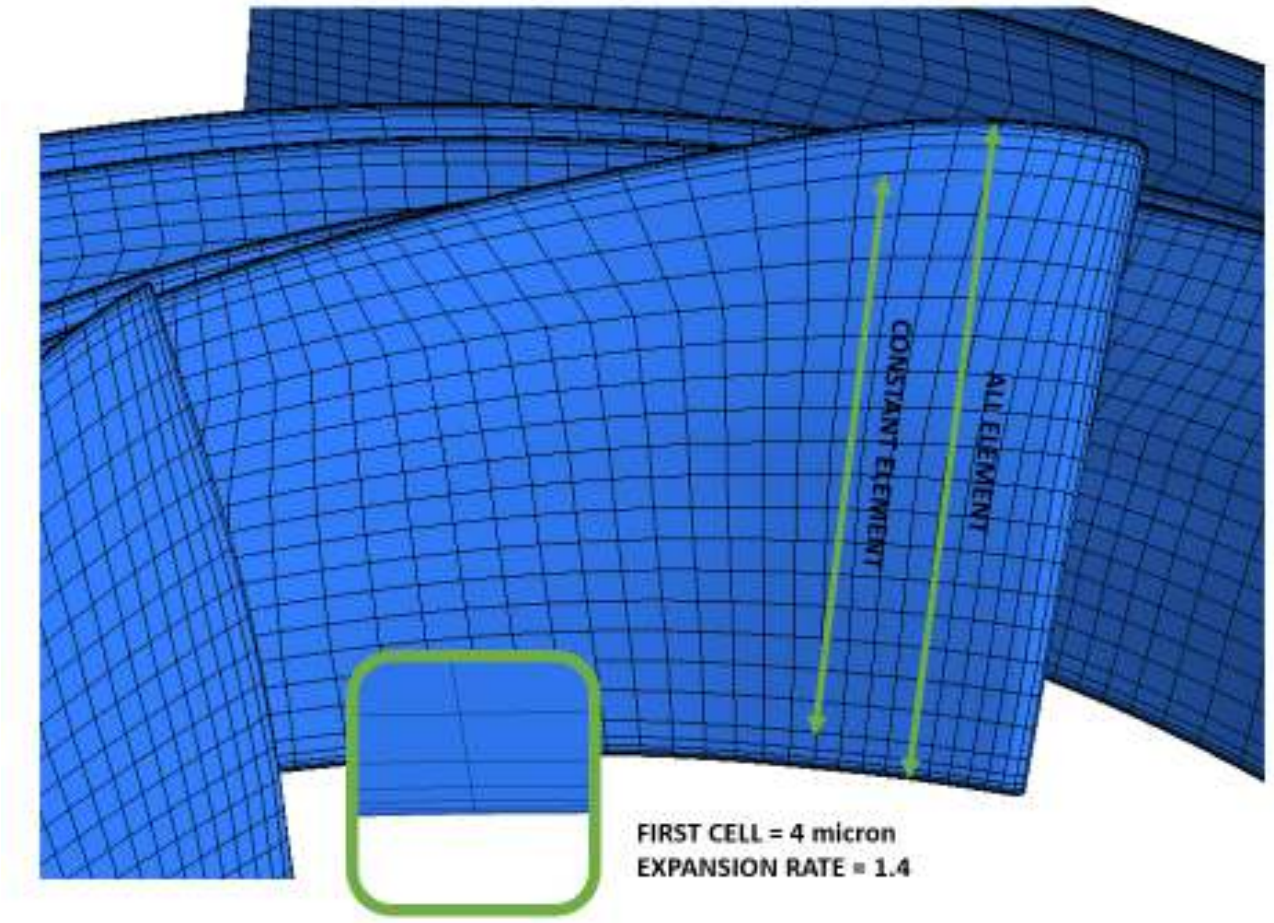

Figure 17: Mesh data and create mesh.

Once that all the field of "Mesh Data" are satisfied, deactivate the command “Suspend Object Updates” and apply “Create Mesh”. The mesh of impeller will be so generated (Figure 18).

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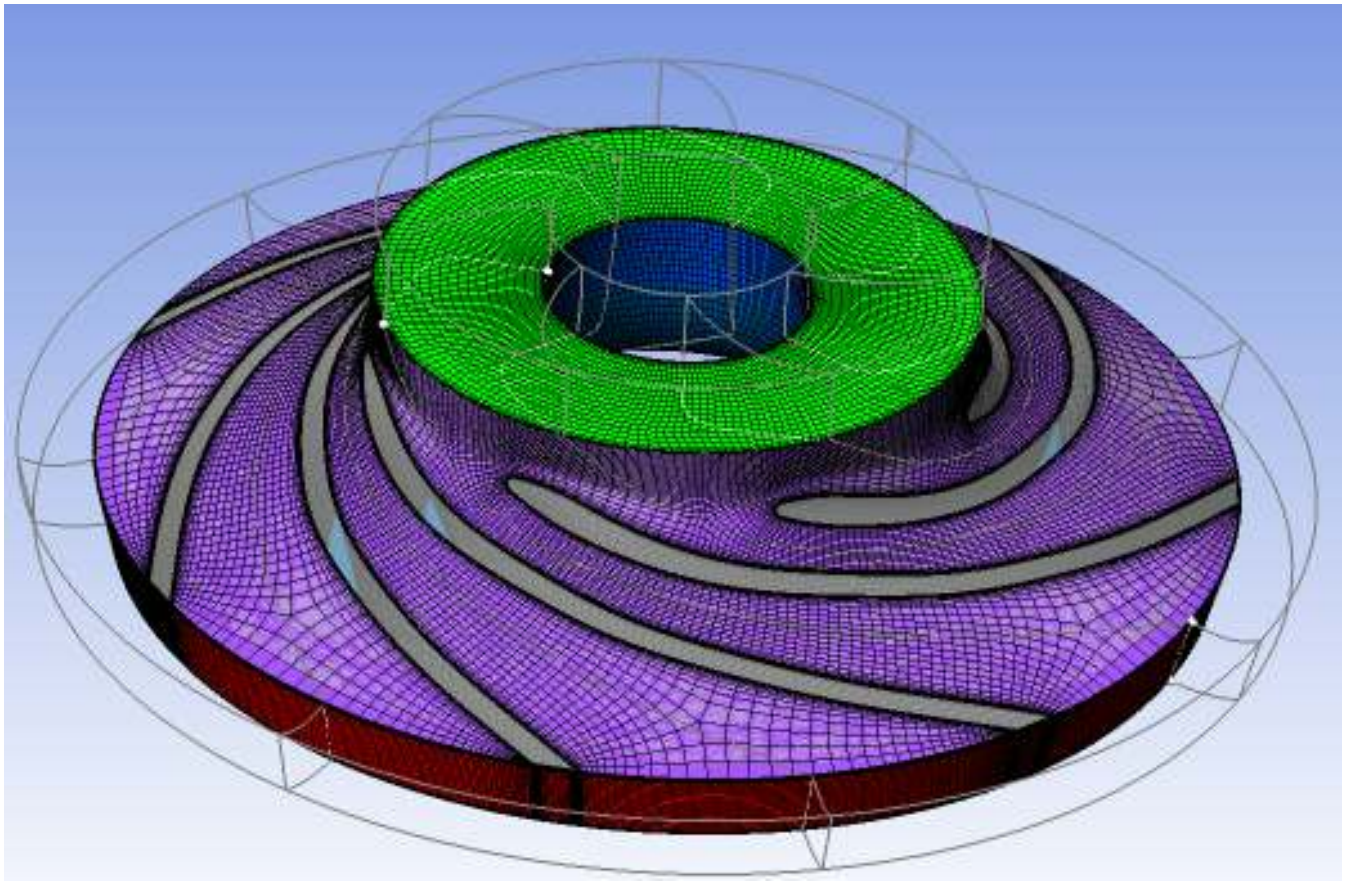

Figure 18: generation mesh impeller.

1.6. Check it out the quality of mesh with the seven parameters of quality available for this tool. View in "Mesh Statistics" the mesh measures of all parameters quality mesh, with color red are marked those parameters that do not respect the "Mesh Limits", in this window is possible to know the correspondents max value and the % elements bad, furthermore is possible view in a graphic also the distribution in % of quality of all elements of mesh (Figure 19).

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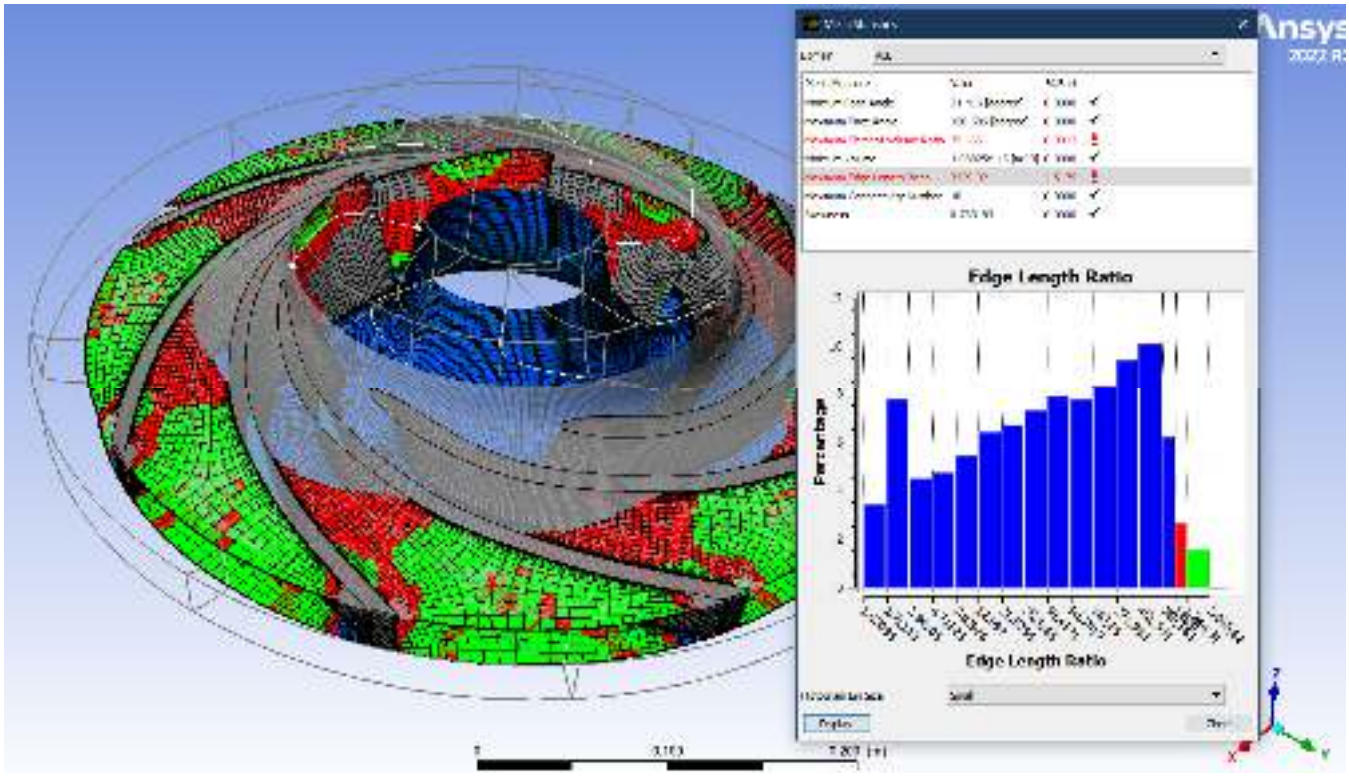

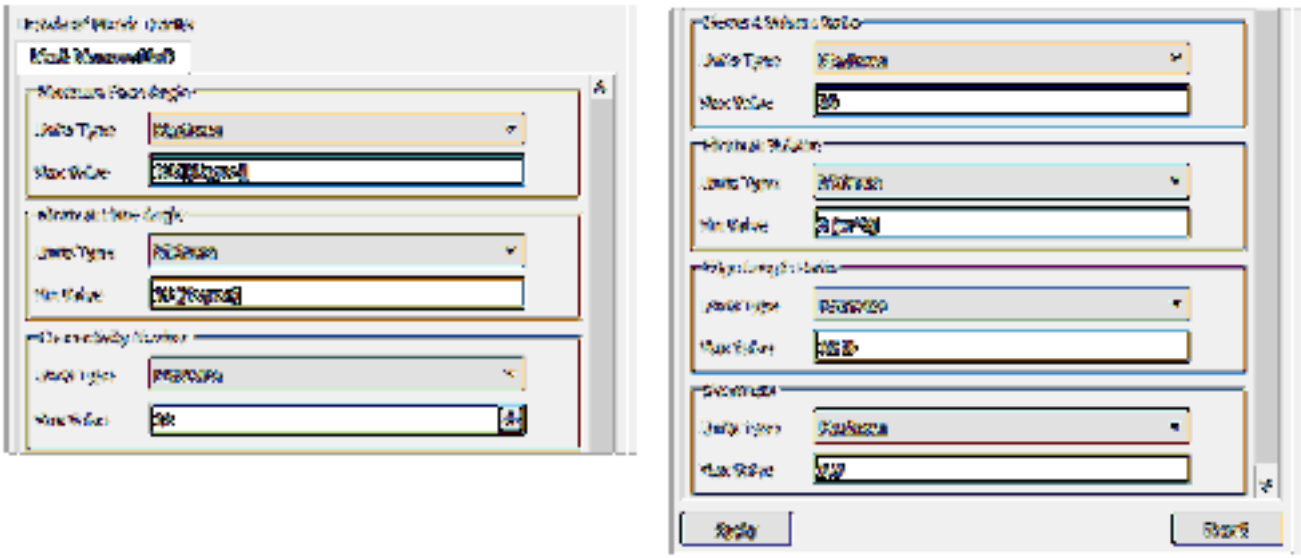

Figure 19: Mesh statistic and Mesh limits.

In general, it is difficult to respect all the limit of parameters but from experience it is necessary and sufficient to get:

- "Minimum Face Angle" ≥ 10°

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- "Minimum Volume" ≥ 0, so with nothing elements negative.

1.7. When the values of "Minimum Face Angle" and "Minimum Volume" are not respected it is necessary improve the quality of mesh.

Mainly the strategies that could be used are the follow:

- Adapt the mesh of the blade with respect to its geometry so that the Leading Edge (LE) is aligned (Figure 20). So, use the selection tool for move and align the control points on the LE. Then is necessary remember of click “apply” and “save & load” for active the changes made

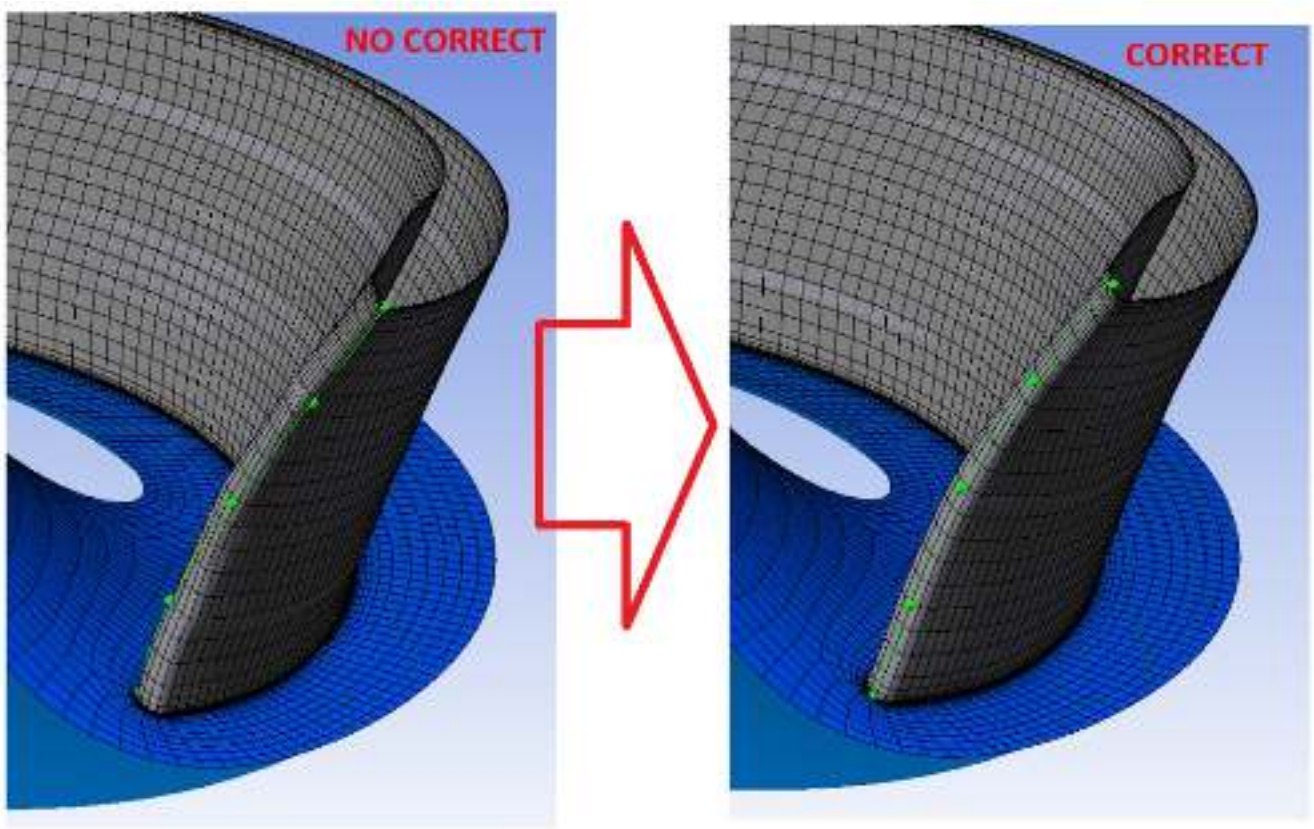

Figure 20: Correct position of mesh on the blade’s geometry.

- Add some "Intermediate Layers" between the hub layer (at span 0.0) and shroud layer (at span 1.0) in proximity to bad elements (Figure 21). In general, it is sufficient to add two or three intermediate layers, to add other layers could not be effective.

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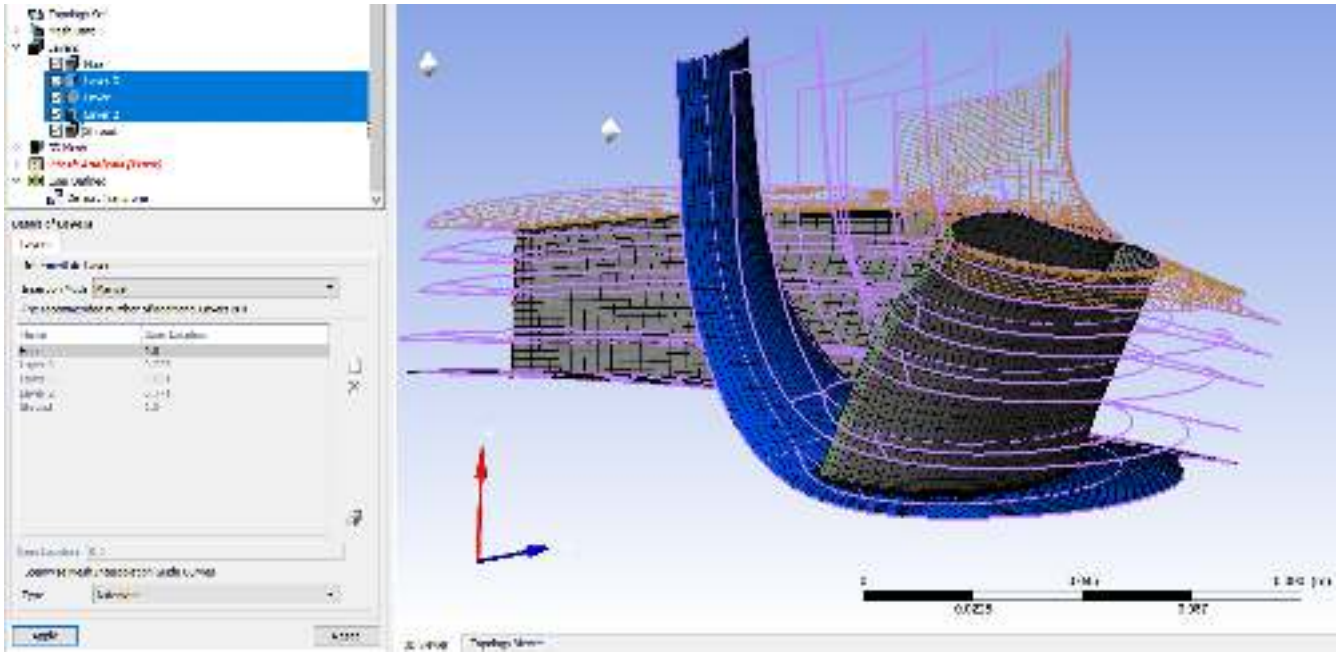

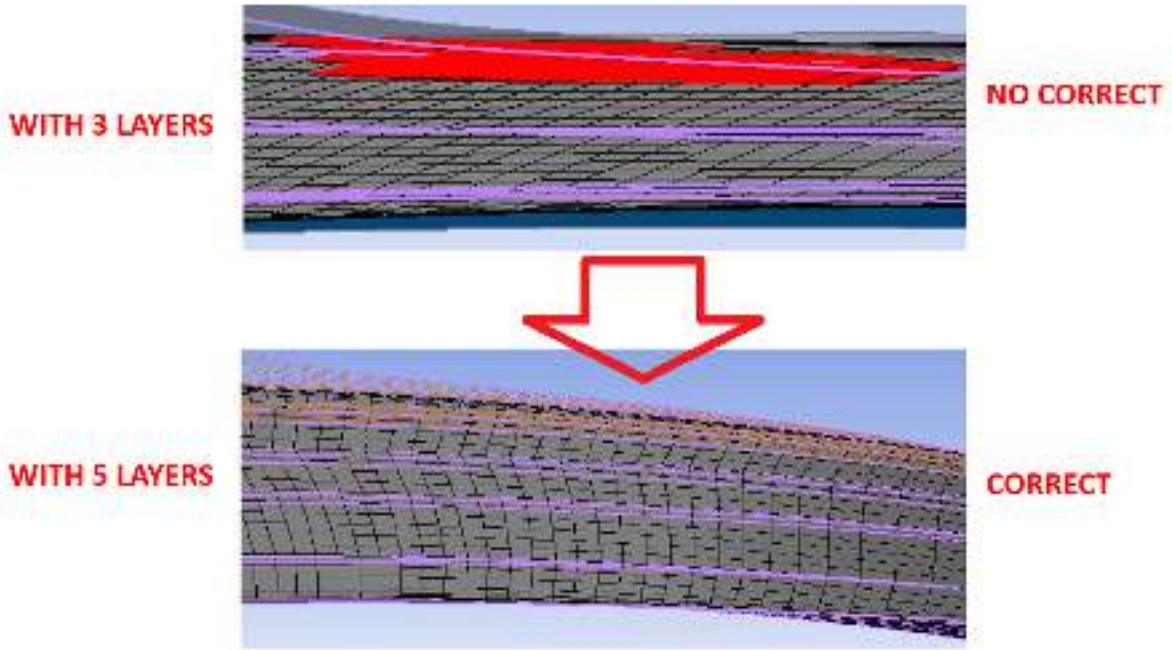

Figure 21: Add intermediate layers for improve quality mesh.

- “Increase Edge Refinement” or “Decrease Edge Refinement” for move and redistribute the cells in the position where the mesh is sparse or full, so lighten up the mesh where the mesh is too full and weigh down where the mesh is too sparse (Figure 22). Note that the total number of element cells is set and has a value constant in “Mesh Data” (for our setting is 200 000 for every vane).

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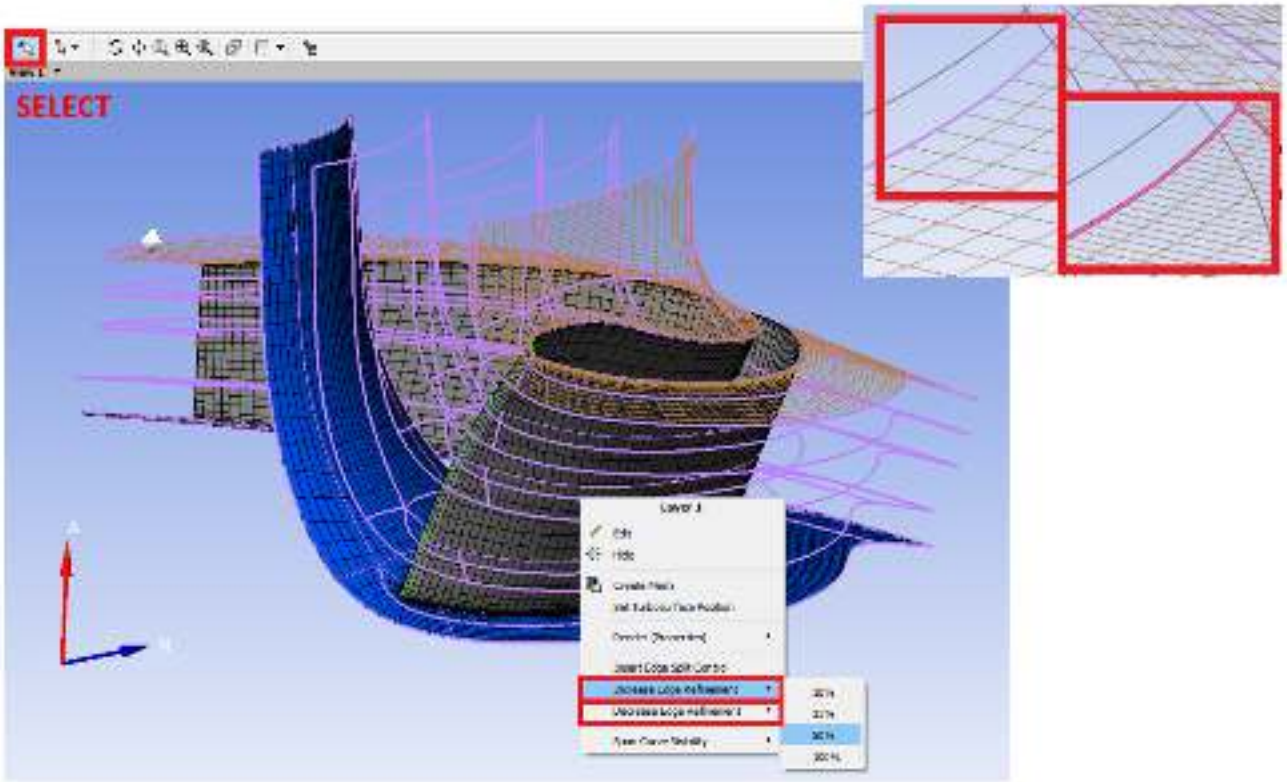

Figure 22: Refinement mesh for improve quality mesh.

- If this first three strategies are not sufficient for improve the quality of mesh is necessary return in “Mesh Data” (operation 3.5) and change the total number of element and the distribution of elements on the spanwise blade.

- if instead the bad quality mesh is located between TE of blade and outer of impeller is necessary increase graduality the value of K until to remove the defective cells.

1.8. When all of criteria are satisfied select "Save State" for save the setting (in format imp_D2_8micro.tst) and select "Save Mesh" for save the mesh (in format imp_001.gtm), Figure 23.

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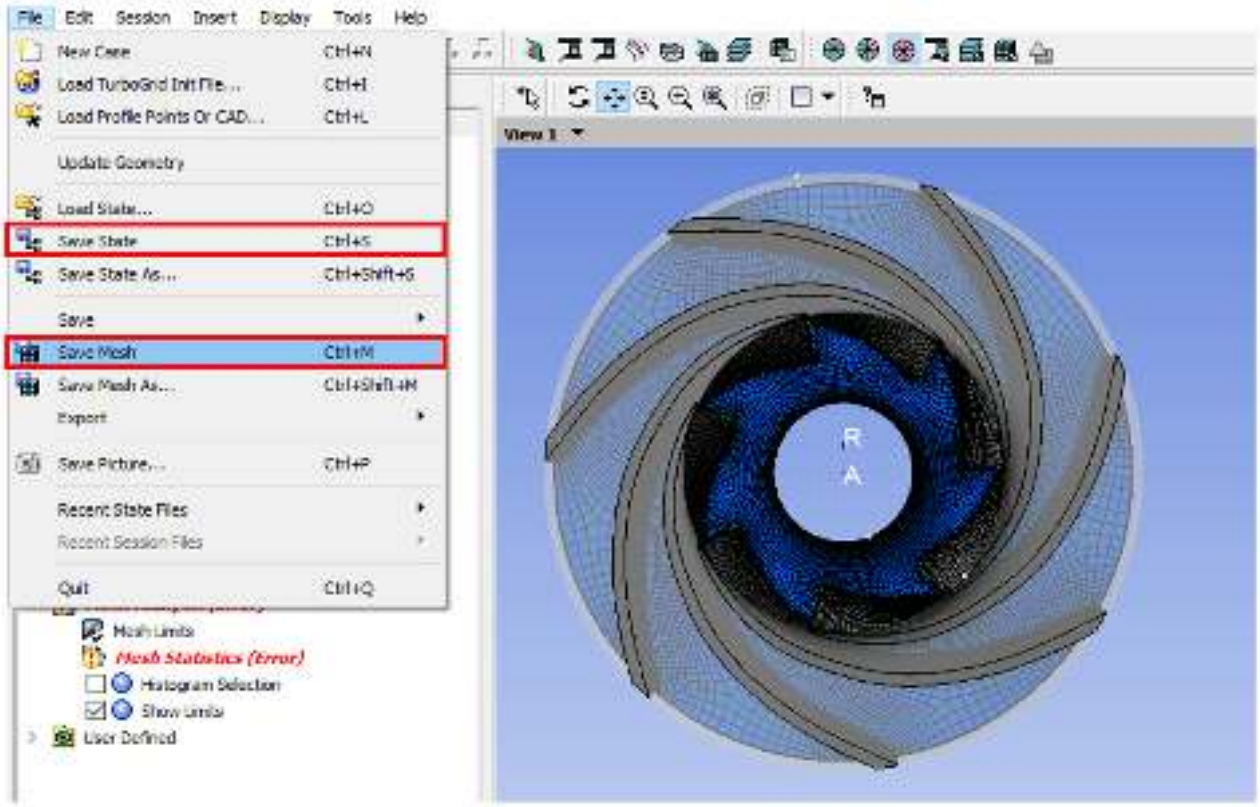

Figure 23: Save state and Save Mesh

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